FUNCTIONAL REQUIREMENTS SPECIFICATION

Banking Application

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1. Introduction

1.1 High-Level User Flow Overview

The following sequence describes the primary path a user takes through the application, from authentication to any functional module:

Authentication Gate → Main Dashboard → Functional Modules → Action Execution → Confirmation/Feedback

- **Step 1 – Authentication:** User arrives at the Login Page. Authenticates via credentials, OAuth, or 2FA. On success, a session token is issued and the user is redirected to the Main Dashboard.
- **Step 2 – Main Dashboard:** Displays a summary of the user's cards, account balances, recent transactions, and quick-action shortcuts. Serves as the navigation hub to all modules.
- **Step 3 – Module Navigation:** User selects a module from the primary navigation (Transfers, Bills, Cards, History, Exchange, Savings & Loans, Investments, Statistics, Support Chat).
- **Step 4 – Action Execution:** Within each module, the user performs specific operations (e.g., initiating a transfer, paying a bill). Sensitive operations require re-authentication or 2FA confirmation.
- **Step 5 – Confirmation:** The system provides immediate feedback (success/failure notification), updates the dashboard state, and logs the action in the transaction history.

2. Authentication and Access Control Module

This module governs all entry points into the application and ensures that only verified users gain access to banking features.

2.1 Login Page (FR-AUTH-001)

2.1.1 Description

The Login Page is the application's entry point. It must present a clean, professional interface with the bank's branding. The page shall support three distinct authentication pathways.

2.1.2 Functional Requirements

FR-AUTH-001.1 – Credential-Based Login

- The system shall display a login form containing two input fields: Email Address (used as the unique username) and Password.
- The Email field shall validate format compliance on the client side before submission.
- The Password field shall mask input by default, with a toggle icon to reveal/hide the password.
- Upon submission, the system shall authenticate the credentials against the user store. On success, issue a session token (JWT or equivalent) and redirect to the Main Dashboard. On failure, display a generic error message ("Invalid email or password") without revealing which field is incorrect.
- After five consecutive failed login attempts from the same IP or for the same account, the system shall temporarily lock the account for 15 minutes and notify the user via their registered email.

FR-AUTH-001.2 – Third-Party OAuth Login

- The Login Page shall display clearly labelled buttons for "Sign in with Google"
  .
- Clicking a third-party button shall initiate an OAuth 2.0 authorization code flow, redirecting the user to the respective provider's consent screen.
- Upon successful OAuth callback, the system shall match the returned email to an existing account or create a new user profile.The user is then redirected to the Main Dashboard.
- If the OAuth provider returns an error or the user denies consent, the system shall display an informative error and return the user to the Login Page.

FR-AUTH-001.3 – Two-Factor Authentication (2FA)

- If the user has enabled 2FA in their profile settings, the system shall, after successful credential verification, redirect to a 2FA Challenge Screen instead of the Dashboard.
- The 2FA Challenge Screen shall prompt the user to enter a 6-digit one-time password (OTP).
- Supported 2FA methods shall include: time-based OTP via an authenticator app (e.g., Google Authenticator, Authy), and OTP delivered via SMS or email to the registered contact.

- The user shall have the option to select their preferred 2FA method if multiple are configured.
- OTP codes shall expire after 60 seconds for TOTP and 5 minutes for SMS/email OTP.
- After three consecutive incorrect OTP entries, the session shall be invalidated and the user must restart the login process from the Login Page.

2.2 Forgot Password Flow (FR-AUTH-002)

2.2.1 Description

This flow allows users who have forgotten their password to securely reset it via a verified email address.

2.2.2 User Flow

- **Step 1:** User clicks "Forgot Password?" link on the Login Page.
- **Step 2:** System displays a form requesting the registered email address.
- **Step 3:** User submits their email. The system always responds with "If an account exists with this email, a reset link has been sent" (to prevent email enumeration).
- **Step 4:** If the email exists, the system generates a one-time, time-limited (30 minutes) password reset token and sends a reset link to the email.
- **Step 5:** User clicks the link, which opens a Set New Password page. The form requires the new password (minimum 8 characters, at least one uppercase letter, one lowercase letter, one digit, and one special character) and a confirmation field.
- **Step 6:** On successful password reset, the system invalidates all existing sessions for the account, displays a confirmation message, and redirects to the Login Page.
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3. Main Dashboard Module

The Main Dashboard is the central hub of the application. It aggregates key account information and provides navigation to all functional modules.

3.1 Dashboard Layout (FR-DASH-001)

3.1.1 Description

Upon successful authentication, the user lands on the Main Dashboard. The dashboard shall be organized into the following zones:

FR-DASH-001.1 – Navigation Bar (persistent across all pages)

- A top or side navigation bar shall be visible on every page of the application.
- Navigation items shall include: Dashboard (Home), Transfers, Bill Payments, Cards, Transaction History, Currency Exchange, Savings & Loans, Investments & Trading, Statistics, Customer Support Chat, and Profile.
- The navigation bar shall highlight the currently active module.
- The bar shall display the user's name or avatar, a notification bell icon with an unread count badge, and a Log Out button.

FR-DASH-001.2 – Cards Overview Panel

- The dashboard shall display all payment cards associated with the user's accounts as interactive card components.
- Each card component shall display: the card brand logo (Visa, Mastercard, etc.), the last four digits of the card number (remaining digits masked as ••••), the cardholder name, the expiry date (MM/YY format), and the associated account name and current balance.
- Card data (full card number, CVV) shall be hidden by default. A "Show Details" action on each card shall require re-authentication (password or biometric) before revealing full card data for a limited time (60 seconds), after which it auto-hides.
- Cards shall be displayed in the user's preferred sort order (see FR-CARD-003 for sorting preferences).

FR-DASH-001.3 – Quick Actions Panel

- A quick actions area shall provide one-click shortcuts to: New Transfer, Pay a Bill, Exchange Currency, and View Full Transaction History.

FR-DASH-001.4 – Recent Transactions Summary

- The dashboard shall display the five most recent transactions across all accounts.
- Each transaction entry shall show: date and time, counterparty name or merchant, amount and currency, transaction type (debit/credit), and a visual indicator for the status (completed, pending, failed).
- A "View All" link shall navigate to the full Transaction History module.

4. User Profile Module

The Profile module enables users to view and manage their personal information and application security settings.

4.1 Personal Information (FR-PROF-001)

- The profile page shall display: full legal name, email address, phone number, date of birth, residential address, and nationality.
- The user shall be able to edit phone number and residential address directly. Changes to the legal name or date of birth shall require a support ticket or KYC re-verification.
- All profile changes shall require password confirmation before saving.

4.2 Security Settings (FR-PROF-002)

- The user shall be able to change their password (requiring current password entry).
- The user shall be able to enable, disable, or reconfigure 2FA methods from this section.
- The user shall be able to manage linked OAuth providers (link/unlink Google, accounts).
- The profile shall display a list of active sessions (device, browser, IP, last active time) with the ability to revoke any individual session.

4.3 Notification Preferences (FR-PROF-003)

- The user shall be able to configure notification channels (push, email, SMS) for: transaction alerts, login alerts, bill payment reminders, and marketing/promotional communications.
- The user shall be able to set a minimum transaction amount threshold for debit alerts.

5. Card Management Module

This module provides full lifecycle management of payment cards associated with the user's accounts.

5.1 View Cards (FR-CARD-001)

- The system shall list all active, frozen, and expired cards belonging to the user.
- Each card entry shall display: card type (debit/credit/prepaid), card brand and last four digits, associated bank account name and IBAN, cardholder name, expiry date, and status (Active, Frozen, Expired, Cancelled).
- The user shall be able to expand a card to view additional details: daily and monthly spending limits, contactless payment status (enabled/disabled), online transaction status (enabled/disabled), and ATM withdrawal limits.

5.2 Show / Hide Sensitive Card Data (FR-CARD-002)

- By default, the full card number and CVV shall be masked.
- The user may trigger "Show Full Card Details" which shall require re-authentication (password entry, biometric, or 2FA code).
- Upon successful re-authentication, the system shall reveal the full 16-digit card number and CVV/CVC for a maximum of 60 seconds, then automatically re-mask the data.
- Each reveal event shall be logged in the user's security audit trail.

5.3 Sort Cards by Preference (FR-CARD-003)

- The user shall be able to reorder cards via drag-and-drop or an explicit "Set as Primary" action.
- The sort preference shall persist across sessions and be reflected on the Main Dashboard.
- The user may sort cards by: custom order (default), balance (ascending/descending), or card type.

5.4 Card Operations (FR-CARD-004)

- **Freeze/Unfreeze Card:** The user shall be able to temporarily freeze a card, instantly blocking all transactions. Unfreezing shall restore normal functionality. Both actions require confirmation.
- **Set Spending Limits:** The user shall be able to modify daily transaction limits, monthly spending caps, ATM withdrawal limits, and contactless payment limits.
- **Enable/Disable Online Transactions:** The user shall be able to toggle online and contactless transaction capabilities per card.
- **Delete (Cancel) Card:** The user shall be able to permanently cancel a card. This action requires 2FA confirmation and a mandatory confirmation dialog explaining the irreversibility. Cancelled cards remain visible in a "Cancelled" state for 90 days for reference, then are archived.
- **Request New Card:** The user shall be able to request a new physical or virtual card, selecting the account to link, card type, and preferred currency.

6. Money Transfer Module

This module enables the user to send money to other accounts, both within the bank and externally.

6.1 New Transfer (FR-TRANS-001)

6.1.1 User Flow

- **Step 1 – Select Source:** User selects the originating account/card from a dropdown of their active accounts, showing the current available balance.
- **Step 2 – Enter Recipient:** User enters the recipient details: recipient name, IBAN or account number, bank name (auto-populated from IBAN where possible), and an optional reference/description.
- **Step 3 – Enter Amount:** User enters the transfer amount and selects the currency. If the currency differs from the source account, the system displays the applicable exchange rate and converted amount in real time.
- **Step 4 – Review:** The system displays a comprehensive summary: source account, recipient details, amount, applicable fees, estimated arrival date, and total debit amount.
- **Step 5 – Confirm:** User confirms the transfer. For amounts exceeding a configurable threshold (e.g., €1,000), the system shall require 2FA confirmation.
- **Step 6 – Result:** System displays a success screen with a transaction reference number, or an error screen with a specific error code and retry option.

6.2 Saved Beneficiaries (FR-TRANS-002)

- The system shall allow the user to save recipient details as a "Beneficiary" for future transfers.
- The user shall be able to view, edit, and delete saved beneficiaries.
- When initiating a new transfer, the user shall be able to select from saved beneficiaries to auto-populate recipient fields.

6.3 Transfer History and Recipients Log (FR-TRANS-003)

- The system shall maintain a log of all recipients the user has sent money to, including date of last transfer, total amount sent, and frequency.
- This log shall be accessible from both the Transfer module and the Transaction History module.
- The user shall be able to initiate a new transfer to any past recipient with one click ("Send Again").

7. Bill Payment Module

This module provides the user with the ability to pay recurring and one-time bills directly from the application.

7.1 Pay a Bill (FR-BILL-001)

7.1.1 User Flow

- **Step 1 – Select Biller:** The user searches for or selects a biller from a categorized directory (Utilities, Telecom, Insurance, Rent, Government, Subscriptions, Other). The directory shall be populated via integration with a bill payment API provider.
- **Step 2 – Enter Details:** User enters the biller-specific reference (e.g., customer/account number), the amount to pay (or selects “Pay in full” if the outstanding balance is retrievable via API), and selects the source account.
- **Step 3 – Review and Confirm:** System displays a summary including biller name, reference, amount, source account, and applicable fees. User confirms the payment.
- **Step 4 – Result:** System processes the payment and displays confirmation with a receipt number. The transaction is recorded in the Transaction History.

7.2 Saved Billers and Recurring Payments (FR-BILL-002)

- The user shall be able to save frequently used billers for quick access.
- The user shall be able to set up recurring automatic payments: selecting a biller, amount (fixed or “full balance”), source account, frequency (weekly, monthly, quarterly), and start/end dates.
- The system shall send a notification 24 hours before each scheduled payment, and a confirmation notification upon execution.
- The user shall be able to view, pause, modify, or cancel any recurring payment.

8. Transaction History Module

This module provides a comprehensive, searchable record of all financial activity across the user's accounts.

8.1 View Transaction History (FR-HIST-001)

- The system shall display a chronological list (newest first by default) of all completed, pending, and failed transactions.
- **Each transaction entry shall display:** date and timestamp, transaction type (transfer, bill payment, card payment, FX exchange, loan disbursement, investment trade, ATM withdrawal, etc.), counterparty or merchant name, amount and currency, debit or credit indicator, running balance after the transaction, and status (Completed, Pending, Failed, Reversed).

8.2 Search, Filter, and Sort (FR-HIST-002)

- **Search:** Free-text search across merchant name, reference number, and description.
- **Filters:** By date range, transaction type, amount range (min/max), specific account or card, status, and debit/credit direction.
- **Sort:** By date (ascending/descending), amount (ascending/descending).
- Filters and sort settings shall be combinable and persist during the session.

8.3 Transaction Details (FR-HIST-003)

- Clicking on any transaction shall open a detail view showing: all fields from the list view, the full originating and destination account/IBAN, any associated fees, the exchange rate applied (for FX transactions), and a downloadable PDF receipt.

8.4 Export (FR-HIST-004)

- The user shall be able to export transaction history for a selected date range in the following formats: PDF statement, CSV file, and Excel (.xlsx) file.

9. Currency Exchange Module

This module enables users to convert funds between different currencies held in their accounts.

9.1 Exchange Currency (FR-FX-001)

9.1.1 User Flow

- **Step 1:** User selects the source account and source currency.
- **Step 2:** User selects the target currency from a list of supported currencies.
- **Step 3:** User enters the amount to exchange (in source or target currency). The system displays the real-time exchange rate (fetched from a rate provider API), the converted amount, any applicable commission or spread, and the final amount the user will receive.
- **Step 4:** User reviews and confirms. The rate shall be locked for 30 seconds from the time of display. If the user does not confirm within 30 seconds, the rate is refreshed.
- **Step 5:** On confirmation, the system debits the source account in the source currency and credits the same or a designated target account in the target currency. A confirmation screen displays the final details and transaction reference.

9.2 Exchange Rate Information (FR-FX-002)

- The module shall display a rate ticker or table showing live exchange rates for popular currency pairs (EUR/USD, EUR/GBP, EUR/RON, etc.).
- The user shall be able to set rate alerts: selecting a currency pair and a target rate, and receiving a push/email notification when the rate is reached.

10. Statistics and Analytics Dashboard

This module provides visual insights into the user's financial activity, enabling better financial decision-making.

10.1 Spending Analytics (FR-STAT-001)

- The dashboard shall display a breakdown of spending by category (e.g., Groceries, Dining, Transport, Entertainment, Bills, Subscriptions) using interactive pie and bar charts.
- The user shall be able to select the time period for analysis: current month, last 3 months, last 6 months, last 12 months, custom date range.
- The system shall automatically categorize transactions based on merchant data. The user shall be able to manually re-categorize individual transactions.

10.2 Income vs. Expenses (FR-STAT-002)

- A line or bar chart shall show monthly income versus expenses over the selected period.
- The dashboard shall display net savings per month and a cumulative savings trend line.

10.3 Account Balance Trends (FR-STAT-003)

- A time-series chart shall show the daily closing balance for each account over the selected period.
- The user shall be able to toggle visibility of individual accounts on the chart.

10.4 Top Recipients and Merchants (FR-STAT-004)

- The dashboard shall display ranked lists of: top 10 merchants by total spend, top 10 transfer recipients by total amount sent, and top billers by total paid.

11. Savings and Loans Module

This module provides access to savings products and loan management features.

11.1 Savings Accounts (FR-SAV-001)

- **View Savings:** The user shall see all active savings accounts with: account name and type (fixed deposit, flexible savings, goal-based savings), current balance and accrued interest, interest rate (annual percentage yield), maturity date (for fixed deposits), and status (Active, Matured, Closed).
- **Open New Savings Account:** The user shall be able to open a new savings product by selecting the savings type, initial deposit amount, target amount and target date (for goal-based savings), funding source account, and deposit frequency (one-time or recurring).
- **Manage Savings:** The user shall be able to make additional deposits, set up or modify automatic deposits, withdraw funds (with any applicable early withdrawal penalties clearly disclosed), and close the savings account.

11.2 Loans (FR-LOAN-001)

- **View Loans:** The user shall see all active and past loans with: loan type (personal, auto, mortgage, student), principal amount, outstanding balance, interest rate, monthly installment amount, next payment due date, remaining term, and repayment progress (percentage paid).
- **Request a New Loan:** The user shall initiate a loan application specifying: loan type, desired amount, preferred term (in months), and purpose. The system shall display an indicative interest rate, monthly installment estimate, and total repayable amount. The application is submitted for review (the outcome is communicated via notification within the application).
- **Pay a Loan Installment:** The user shall be able to make a manual payment toward any active loan, selecting the source account and amount (minimum installment or a custom amount for early repayment). The system shall show the impact on remaining balance and term.
- **Loan Repayment Schedule:** The user shall be able to view and download the full amortization schedule for any active loan, showing each installment's date, principal portion, interest portion, and remaining balance.

12. Investments and Trading Module

This module enables users to invest in financial instruments and trade cryptocurrencies directly from the application.

12.1 Portfolio Overview (FR-INV-001)

- The system shall display the user's investment portfolio with: total portfolio value, total gain/loss (absolute and percentage), and individual asset holdings categorized by type (Stocks, ETFs, Bonds, Cryptocurrency, Other).
- Each asset entry shall display: asset name and ticker symbol, quantity held, average purchase price, current market price (updated in real time or near-real time), and unrealized gain/loss.

12.3 Cryptocurrency Trading (FR-INV-003)

- The cryptocurrency section shall support viewing and trading major cryptocurrencies (e.g., BTC, ETH, SOL, ADA, and other configurable tokens).
- The user flow for buying and selling cryptocurrency, with the addition of: a crypto-specific wallet view showing token balances, network fee estimation displayed before confirmation, and a notice regarding the volatility and risk associated with cryptocurrency investments.

12.4 Transaction Log and Statements (FR-INV-004)

- All investment and trading transactions shall be logged with: date and time, asset, action (Buy/Sell), quantity, price per unit, total value, fees, and order type (Market/Limit).
- The user shall be able to export investment statements for a given period as a PDF or CSV.

13. Customer Support Chat Module

This module provides real-time communication between the user and the bank's customer support team.

13.1 Live Chat (FR-CHAT-001)

- The application shall provide a dedicated, in-app chat interface accessible from the navigation menu and via a persistent floating chat icon.
- Upon opening the chat, the system shall display: a welcome message with estimated wait time, an option to describe the issue category (Account, Cards, Transfers, Loans, Technical Issue, Other) before connecting to an agent, and full chat history from previous support sessions (persisted per user).

13.2 Chat Functionality (FR-CHAT-002)

- **Messaging:** Real-time text messaging between the user and a support agent.
- **File Sharing:** The user shall be able to attach and send images and PDF documents (up to 10 MB per file) for supporting evidence or documentation.
- **Chat Transcript:** At the end of each session, the user shall be offered the option to receive a copy of the chat transcript via email.
- **Feedback:** After a chat session ends, the user shall be prompted to rate the support experience (1–5 stars) and provide optional written feedback.

13.3 Automated Support (FR-CHAT-003)

- Before connecting to a live agent, the system may present an AI chatbot or FAQ-based assistant that attempts to resolve common queries (balance inquiries, transaction status, card freeze, etc.).
- If the chatbot cannot resolve the issue, the conversation shall be seamlessly escalated to a human agent, preserving the full conversation context.

15. Cross-Cutting Functional Constraints

The following functional constraints apply across all modules and should be considered during architectural design and implementation.

15.1 Responsive Design (FR-SYS-001)

- All screens and modules shall be fully responsive, functioning correctly on desktop (1920×1080 and above), tablet (768×1024), and mobile (375×812) viewports.
- Critical actions (login, transfers, bill pay) shall be fully operable on mobile without horizontal scrolling.

15.2 Accessibility (FR-SYS-002)

- The application shall comply with WCAG 2.1 Level AA guidelines.
- All interactive elements shall be keyboard-navigable. All images and icons shall have appropriate alt text. Form fields shall have associated labels.

15.3 Localization (FR-SYS-003)

- The application shall support at least two languages (e.g., English and Romanian) with a language selector in the profile or navigation.
- All dates, times, and currency values shall be formatted according to the user's locale.

15.4 Error Handling (FR-SYS-004)

- All errors shall display user-friendly messages with a suggested corrective action.
- Network errors shall trigger automatic retry (up to 3 attempts) with exponential backoff, followed by an offline-mode notification.
- Critical transaction errors shall never result in silent failures; the user shall always receive explicit success or failure confirmation.
-